

FORESTBURG AGCM, AB, Canada

WMO: 716740

Lat: 52.5485N

Lon: 112.1234W

Elev: 2365

StdP: 13.48

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.3	-18.0	-29.8	1.1	-23.0	-24.2	1.6	-17.8	25.6	29.7	23.8	33.0	5.3	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	84.3	63.5	81.0	61.7	78.0	60.5	67.0	79.1	64.8	76.6	62.9	74.0	8.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	92.7	73.5	60.4	85.5	70.5	58.4	79.7	67.9	32.9	79.2	31.1	77.6	29.6	73.7	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	19.9	17.7	DB	-31.1	90.1	4.5	4.2	-34.3	93.1	-36.9	95.6	-39.4	98.0	-42.7	101.1	(4)
(5)			WB	-31.3	71.0	4.4	2.5	-34.5	72.7	-37.1	74.2	-39.5	75.6	-42.8	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	14.4	12.3	23.3	37.5	51.5	58.9	62.7	60.5	51.7	39.8	24.3	12.1		
	DBStd	21.56	14.91	15.43	14.22	9.60	7.69	5.16	5.16	5.48	7.96	8.85	12.82	13.65		
	HDD50	5800	1102	1054	829	382	76	3	0	1	74	333	772	1174		
	HDD65	10086	1567	1474	1294	824	422	195	108	157	402	782	1222	1639		
	CDD50	1262	0	0	0	9	121	269	393	328	125	16	1	0		
	CDD65	71	0	0	0	0	2	11	36	19	3	0	0	0		
	CDH74	1698	0	0	0	8	152	209	629	515	179	6	0	0		
	CDH80	394	0	0	0	0	23	29	173	125	44	0	0	0		
	WSAvg	8.4	8.7	8.5	8.1	9.3	8.8	9.0	7.3	7.5	7.7	9.6	8.2	8.4		
	PrecAvg	16.2	0.6	0.5	0.6	0.7	1.7	3.1	3.1	2.3	1.7	0.6	0.5	0.6		
(15) Precipitation	PrecMax	20.2	1.7	1.4	1.8	1.8	4.7	6.2	6.0	5.7	4.7	1.7	1.2	1.7		
	PrecMin	12.6	0.1	0.0	0.0	0.0	0.4	0.9	0.9	0.4	0.0	0.0	0.0	0.1		
	PrecStd	2.4	0.4	0.4	0.4	0.5	1.1	1.7	1.4	1.4	1.2	0.4	0.3	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	45.3	56.1	72.8	83.0	83.2	88.8	87.6	85.7	73.9	57.9	43.8		
		MCWB	37.8	38.4	44.7	49.5	55.8	61.5	66.4	62.8	59.9	54.6	45.2	38.1		
	2%	DB	41.3	40.4	50.2	65.1	78.4	79.6	84.6	83.4	79.0	65.3	49.8	39.5		
		MCWB	35.8	34.8	41.5	46.8	54.6	60.9	66.6	62.6	57.3	50.0	40.9	34.3		
	5%	DB	38.3	37.0	44.0	59.1	73.5	75.9	80.9	79.7	74.3	59.6	44.1	35.1		
		MCWB	33.9	32.7	37.4	44.2	52.9	59.5	64.6	61.8	56.2	46.9	37.1	30.8		
	10%	DB	35.0	33.7	40.0	54.1	68.4	72.1	76.9	76.1	69.0	54.6	40.4	31.2		
		MCWB	31.0	30.6	34.8	42.1	50.6	57.5	63.4	60.8	54.1	44.2	34.8	28.0		
	0.4%	WB	38.2	38.6	46.6	51.2	58.1	66.3	71.5	68.2	61.9	55.5	46.6	38.6		
		MCDB	43.4	44.5	55.7	71.8	77.8	78.1	82.6	79.3	78.7	71.2	58.6	43.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.1	35.2	41.2	47.4	55.8	63.2	68.2	65.7	59.0	51.4	41.5	34.4		
		MCDB	40.7	39.7	49.0	63.3	73.6	75.0	80.9	78.2	75.4	61.6	48.1	39.1		
	5%	WB	34.1	33.0	37.8	45.0	54.0	61.2	66.1	63.6	56.7	48.1	37.9	31.2		
		MCDB	38.4	36.5	43.2	57.6	70.1	71.7	77.6	76.0	71.3	58.4	43.1	34.9		
	10%	WB	31.3	30.7	35.0	42.6	52.1	59.0	64.1	61.5	54.9	45.1	35.4	27.9		
		MCDB	34.9	34.0	39.5	53.1	66.3	68.9	75.1	73.2	68.1	54.0	39.4	31.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	18.9	18.3	21.4	25.2	22.0	24.4	25.6	25.2	22.1	17.8	16.5		
		MCDBR	18.3	21.2	22.4	32.0	33.4	28.5	31.5	33.8	36.3	31.0	22.4	19.1		
		MCWBR	14.2	16.7	15.0	16.6	14.3	12.8	15.6	15.8	16.7	17.3	14.7	15.1		
		MCDBR	18.9	19.9	21.1	29.0	29.5	24.0	28.3	29.8	32.6	29.0	20.9	19.0		
	5% WB	MCWBR	15.1	15.9	14.5	15.5	13.2	12.2	15.2	15.2	15.7	16.5	14.5	15.4		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.283	0.293	0.331	0.352	0.364	0.358	0.372	0.317	0.284	0.258	0.247		
	taud		2.276	2.320	2.413	2.361	2.351	2.350	2.400	2.341	2.460	2.530	2.457	2.319		
	Ebn at Noon		232	266	288	287	285	281	281	270	276	266	242	222		
	Edh at Noon		20	25	29	35	37	37	35	35	27	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		344	677	1131	1500	1828	1914	2010	1639	1144	682	346	253		
	RadStd		31	55	83	143	119	127	103	120	122	84	36	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (86 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page